

UNIVERSITY NAME, SPRING 2026
MATH 100 - COURSE NAME

Title of this Assignment

INSTRUCTOR:	INSTRUCTOR NAME
STUDENT:	STUDENT NAME
DUE DATE:	MMM DD, YYYY

EXERCISE 1. Let $A \subset [0, 1]$ be the union of open intervals (a_i, b_i) such that each rational number in $(0, 1)$ is contained in some interval (a_i, b_i) . Show that $\text{Bd } A = [0, 1] \setminus A$. Give an example how to construct such intervals such that $\sum_{i=1}^{\infty} (b_i - a_i) < 1$. Explain why in this case A , even though it is open, it is not Jordan Measurable, so χ_A is not integrable.

SOLUTION. The set A contains all rationals in $(0, 1)$, which are dense in $[0, 1]$. Thus every neighborhood of any point in $[0, 1]$ intersects A (indeed any neighborhood contains a rational $q \in (0, 1)$ and by assumption that rational lies in some (a_i, b_i)), so $\overline{A} = [0, 1]$. Since A is a union of open intervals, $\text{Int } A = A$. Therefore,

$$\partial A = \overline{A} \setminus \text{Int } A = [0, 1] \setminus A.$$

For the example, enumerate the rationals in $(0, 1)$ as $q_1, q_2, q_3, \dots$. For each i take an open interval centered at q_i of length $2^{-(i+1)}$, for instance

$$(a_i, b_i) = \left(q_i - \frac{1}{2^{i+2}}, q_i + \frac{1}{2^{i+2}} \right),$$

and if necessary replace (a_i, b_i) by $(a_i, b_i) \cap (0, 1)$ to keep it inside $(0, 1)$. Then

$$b_i - a_i = \frac{1}{2^{i+1}}, \quad \sum_{i=1}^{\infty} (b_i - a_i) = \sum_{i=1}^{\infty} \frac{1}{2^{i+1}} = \frac{1}{2} < 1,$$

and by construction every rational in $(0, 1)$ lies in some (a_i, b_i) . Finally, to see A is not Jordan measurable: since $A = \bigcup_i (a_i, b_i)$ and the total length of these intervals is $\sum_i (b_i - a_i) < 1$, the complement $[0, 1] \setminus A = \partial A$ has length at least $1 - \sum_i (b_i - a_i) > 0$. In particular ∂A does not have measure zero (Jordan content zero), so A is not Jordan measurable. By the usual criterion (a bounded set is Jordan measurable iff its characteristic function is Riemann integrable) the characteristic function χ_A is therefore not integrable on $[0, 1]$. $\square$